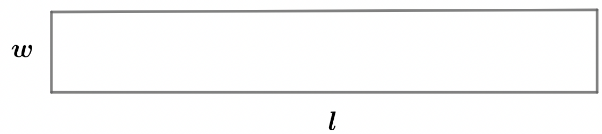
# 2022B F=ma Exam: Problem 21

Kevin S. Huang




We have

$$w = 1\,\text{cm}, \ \frac{\Delta w}{w} = 0.01$$
$$l = 100\,\text{cm}, \ \frac{\Delta l}{l} = 0.001$$

The area of the rectangle is

$$A = wl = 100\,\text{cm}^2$$

Recall the relative uncertainties add in quadrature when two independent variables are multiplied together:

$$\frac{\Delta A}{A} = \sqrt{\left(\frac{\Delta w}{w}\right)^2 + \left(\frac{\Delta l}{l}\right)^2} = \sqrt{0.01^2 + 0.001^2} \approx 0.01 = 1\%$$

The perimeter of the rectangle is

$$p = 2(w + l)$$

Recall the absolute uncertainties add in quadrature when two independent variables are added together:

$$\Delta p = 2\sqrt{(\Delta w)^2 + (\Delta l)^2} = 2\sqrt{(0.01\,\text{cm})^2 + (0.1\,\text{cm})^2} \approx 0.2\,\text{cm}$$

The relative uncertainty is

$$\frac{\Delta p}{p} = \frac{0.2\,\text{cm}}{2(100\,\text{cm} + 1\,\text{cm})} \approx 0.1\%$$

so the answer is $\boxed{\text{B}}$.